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3.3.1 Number of research papers published per teacher in the journals notified on UGC Care list during the last five years

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List of paper publications

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S.NO	Title of the paper	Name of the author/s	Name of the journal	Page No
1	Physicochemical, pharmacokinetic and pharmacodynamic characterization of isradipine tablets for controlled release	D.N.Venkatesh, S.N.Meyyanathan, R. Shanmughan, S.S.Kamatham, J.R.Campos, J.Dias Ferreira, E. Sanchez-lopez, J.C.Cardoso, P. Severino, E.B.Souto	Pharmaceutical technology and development	3-4



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Physicochemical, pharmacokinetic, and pharmacodynamic characterization of isradipine tablets for controlled release

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ABSTRACT

Isradipine is a dihydropyridine calcium channel blocker (CCB) commonly used as vasodilator with antihypertensive properties. A remote-controlled release formulation for isradipine would substantially improve the clinical outcomes of the patients requiring chronic long-term treatment. In this work, sustained release (SR) tablets of isradipine, composed of hydroxypropylmethyl cellulose (HPMC), have been produced by wet granulation and their *in vitro* and *in vivo* characterization was compared to a conventional tablet dosage form of immediate release (IR) as preliminary assessment. Tablets composed of 15.0% (wt/wt) HPMC exhibited a SR profile over a period of 24 hours. The release of isradipine followed a Fickian diffusion pattern obeying to the first order kinetics and the extent of absorption was even higher in comparison to the developed conventional tablets, which showed immediate drug release. *In vivo* studies were carried out in rabbits, showing that the extent of isradipine absorption from the developed tablets was higher in comparison to IR tablets due to the modified release profile obtained for the former. **100% bioequivalence** results suggest that SR tablets of isradipine are an efficient solid dosage form to overcome the limitations encountered in conventional IR tablets.

Isradipine is a well-known poorly water-soluble antihypertensive drug, commonly used in the treatment and management of hypertension. Its posology is usually 2.5 mg for immediate-release capsules, twice a day, because of its reduced water solubility (<10 µg/L at 37°C) and therefore short plasma half-life (Hof and Rüegg 1988; Johnson et al. 2005; Ilijic et al. 2011; Tran et al. 2013). Sustained release (SR) solid dosage forms for oral administration may be effective to improve the therapeutic outcomes with drug molecules that exhibit a narrow therapeutic range and/or have a fast clearance from the blood (Venkatesh et al. 2020a, 2020b). Several categories of modified-release formulations for oral administration have been developed to improve the clinical outcomes of poorly water-soluble drugs and drugs with short half-lives, and also to increase patients' compliance (Varum et al. 2010; Sahu et al. 2017). Modified-release formulations (e.g. controlled, sustained, delayed, and prolonged) aim at delivering drugs at a predetermined rate. These dosage forms are characterized by a release kinetics that keeps the drug concentration within its therapeutic window for a pre-determined time-frame (Yuen 2010;

KEYWORDS

Isradipine; sustained release; immediate release; hydrophilic matrix; bioavailability evaluation; pharmacokinetics

Goyanes et al. 2015). Besides the marketed 5 mg Dynacirc[®] capsules recommended for the treatment of hypertension (Schran et al. 1988), isradipine SR tablets have not yet been proposed for the treatment of left ventricular morphology function in systemic hypertension, or to minimize cardiovascular risk factors occurring in smoking cessation and type II diabetes mellitus (Staessen et al. 1988; Van Den Berg and Dehmer 1988; Grossman et al. 1991; Kumar et al. 2016). Its very limited water-solubility and short biological half-life, together with the need to increase patients' compliance, especially in the management of hypertension, demand for new oral SR formulations for this drug. The novelty of this work is the development of a modified release tablet formulation that may reduce the number of doses required for the treatment of hypertension, thereby reducing the risk of overdose and associated adverse side effects.

Due to their versatility, biocompatibility, low cost, and regulatory acceptance, hydrophilic polymeric matrices have been commonly used in oral drug delivery to obtain a desirable release profile.



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